

SIOC 217C: Climate and Climate Change

Homework #1

Consider an atmosphere filled by an absorber with uniform mixing ratio that absorbs equally at all terrestrial wavelengths. The optical thickness for this absorber is zero at the top of the atmosphere and has value τ_0 at the surface. For simplicity, assume that atmospheric pressure and density vary exponentially with constant scale height H and ignore any variation of scale height with temperature. Assume the surface is a blackbody and the atmosphere is a graybody (e.g., emits like a blackbody scaled by a fraction equal to emissivity) at terrestrial wavelengths. Also assume that optical thickness can be multiplied by a constant “diffusivity factor” r in order to take into account radiation goes at all angles of the hemisphere and not just in the vertical direction. For this assignment, it will be most convenient to create a model with layers that have equal thickness in height coordinates.

1. a) Let $\tau(z)$ be the value of optical thickness at height z integrated downward from the top of the atmosphere, with a value of τ_0 at the surface. Provide an expression for $\tau(z)$ as a function of τ_0 , z , and any other necessary parameters.
b) On one plot, show how $\tau(z)$ varies with z in the range $0 \leq z \leq 5H$ for $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$.
2. a) Let $\mathcal{T}(z, \infty)$ be the fraction of broadband flux (irradiance) emitted upward by a thin layer of the atmosphere at height z that is transmitted to space. Provide an expression for $\mathcal{T}(z, \infty)$ as a function of $\tau(z)$, r , and any other necessary parameters.
b) On one plot, show how $\mathcal{T}(z, \infty)$ varies with z in the range $0 \leq z \leq 5H$ for $r = 1.66$ and $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$.
3. a) Let B_0 be the amount of broadband hemispheric flux (irradiance) emitted upward by the surface, which can be approximated as a blackbody at terrestrial wavelengths. Let surface temperature be T_0 . Provide an expression for B_0 as a function of T_0 and any other necessary parameters.
b) Let $\mathcal{B}_0(\infty)$ be the amount of broadband flux (irradiance) emitted upward by the surface that is transmitted to space. Provide an expression for $\mathcal{B}_0(\infty)$ as a function of T_0 , $\tau(z)$, r , and any other necessary parameters.
c) Calculate values of $\mathcal{B}_0(\infty) / \sigma T_0^4$ for $r = 1.66$ and $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$.
4. a) Consider a thin layer of the atmosphere at height z that has optical thickness $\Delta\tau$, where $\Delta\tau \ll 1$. Let $\mathcal{A}(z)$ be the fraction of broadband flux (irradiance) upwelling from below directly into the layer that is absorbed (the absorptivity of the layer). Provide an expression for $\mathcal{A}(z)$ as a function of $\Delta\tau$, r , and any other necessary parameters as the leading term of a Taylor series.

- b) Let Δz be the geometrical thickness of the thin atmospheric layer, where $\Delta z / H \ll 1$. Provide an expression for $\mathcal{A}(z)$ as a function of Δz , $\tau(z)$, r , and any other necessary parameters as the leading term of a Taylor series.
- c) On one plot, show how $\mathcal{A}(z)$ varies with z in the range $0 \leq z \leq 5H$ for $r = 1.66$ and $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$. Specify what value of $\Delta z / H$ you used.
5. a) Let $B(z)$ be the amount of broadband hemispheric flux (irradiance) emitted upward by a thin atmospheric layer at height z with geometric thickness Δz , where $\Delta z / H \ll 1$. For simplicity, assume atmospheric temperature T_0 is uniform. Let the emissivity of the layer be $\varepsilon(z)$. Provide an expression for $B(z)$ as a function of T_0 , atmospheric emissivity $\varepsilon(z)$, and any other necessary parameters.
- b) Provide an expression for atmospheric emissivity $\varepsilon(z)$ as a function of Δz , $\tau(z)$, r , and any other necessary parameters as the leading term of a Taylor series.
- c) Let $\mathcal{B}(z, \infty)$ be the amount of broadband flux (irradiance) emitted upward by a thin atmospheric layer at height z that is transmitted to space. Provide an expression for $\mathcal{B}(z, \infty)$ as a function of T_0 , Δz , $\tau(z)$, r , and any other necessary parameters.
- d) Provide an expression for the value of $\tau_{\max}(z)$ at which the amount of broadband flux emitted by a thin atmospheric layer that is transmitted to space is maximum. Assume that Δz is constant.
- e) Provide an expression for the value of $z_{\max} / H$ at which the amount of broadband flux emitted by a thin atmospheric layer that is transmitted to space is maximum.
- f) Let $\mathcal{T}(z_{\max}, \infty)$ be the fraction of broadband flux (irradiance) emitted upward by a thin layer of the atmosphere at height $z_{\max}$ that is transmitted to space. Provide an expression for $\mathcal{T}(z_{\max}, \infty)$ as a function of any necessary parameters.
- g) Let $\varepsilon(z_{\max})$ be the atmospheric emissivity at height $z_{\max}$. Provide an expression for $\varepsilon(z_{\max})$ as a function of Δz and any other necessary parameters.
- h) Calculate values of $z_{\max} / H$ for $r = 1.66$ and $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$. Note that if $r \tau_0 \leq 1$, maximum emission to space occurs at the surface.
- i) On one plot, show how the ratio $\mathcal{B}(z, \infty) / \sigma T_0^4$ varies with z in the range $0 \leq z \leq 5H$ for $r = 1.66$ and $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$. Specify what value of $\Delta z / H$ you used. Verify that your various calculations and plots are consistent.
6. Using the model for the atmosphere developed for the previous questions, consider what will immediately happen if the absorber mixing ratio in the atmosphere suddenly increases.
- a) Will τ_0 increase, decrease, or remain the same?
- b) Will $\tau_{\max}(z)$ increase, decrease, or remain the same?
- c) Will $z_{\max} / H$ increase, decrease, or remain the same?
- d) Will $\mathcal{T}(z_{\max}, \infty)$ increase, decrease, or remain the same?

- e) Will the fraction of broadband flux emitted by the surface and transmitted to space increase, decrease, or remain the same?
 - f) Will the fraction of broadband flux emitted at height $z_{\max}$ and transmitted to space increase, decrease, or remain the same?
 - g) Will the emissivity of a thin atmospheric layer at height $z_{\max}$ increase, decrease, or remain the same?
 - h) Treating a thin atmospheric layer at height $z_{\max}$ as a blackbody, will the amount of broadband flux emitted at height $z_{\max}$ increase, decrease, or remain the same if atmospheric temperature monotonically decreases with height above the surface?
 - i) Treating a thin atmospheric layer at height $z_{\max}$ as a blackbody, will the amount of broadband flux emitted at height $z_{\max}$ increase, decrease, or remain the same if atmospheric temperature monotonically increases with height above the surface?
 - j) Under what condition will the greenhouse effect of an atmosphere become weaker when absorber mixing ratio increases?
7. a) Let $\mathcal{T}(z,0)$ be the fraction of broadband flux (irradiance) emitted downward by a thin layer of the atmosphere at height z that is transmitted to the surface. Provide an expression for $\mathcal{T}(z,0)$ as a function of $\tau(z)$, r , and any other necessary parameters.
- b) On one plot, show how $\mathcal{T}(z,0)$ varies with z in the range $0 \leq z \leq 5H$ for $r = 1.66$ and $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$.
8. a) Let $\mathcal{B}(z,0)$ be the amount of broadband flux (irradiance) emitted downward by a thin atmospheric layer at height z that is transmitted to the surface. For this problem, assume that atmospheric temperature T_0 is uniform. Provide an expression for $\mathcal{B}(z,0)$ as a function of $\mathcal{T}(z,0)$, T_0 , $\varepsilon(z)$, and any other necessary parameters.
- b) On one plot, show how the ratio $\mathcal{B}(z,0) / \sigma T_0^4$ varies with z in the range $0 \leq z \leq 5H$ for $r = 1.66$ and $\tau_0 = 0.1$, $\tau_0 = 0.3$, $\tau_0 = 1$, and $\tau_0 = 3$. Specify what value of $\Delta z / H$ you used. Note that the vertical profile of emission to the surface does not vary monotonically with τ_0 . If you do not understand why this is, come talk to me.

We will now apply the model to investigate differences in radiation flux between an atmosphere with $1\times\text{CO}_2$ concentration and an atmosphere with $2\times\text{CO}_2$ concentration. In both cases, we will assume that surface temperature $T_s = 288\text{ K}$ and that temperature varies linearly with height according to $T(z) = T_s - \gamma z / H$. If scale height H has a constant value of 8 km and lapse rate has a constant value of 6.5 K km^{-1} , then $\gamma = 52\text{ K}$. Since the model atmosphere has the same optical thickness for all wavelengths, it is not possible to match all radiative characteristics of Earth's atmosphere, but we can match the observed average height of maximum emission to space by choosing a value of $\tau_0 = 2.2$ for $1\times\text{CO}_2$ concentration. Choosing a value of $\tau_0 = 2.3$ for $2\times\text{CO}_2$ concentration will result in changes in radiation flux similar to those derived from more sophisticated models.

9. a) Let $\mathcal{B}(z,0)$ be the amount of flux emitted downward by a thin atmospheric layer at height z that is transmitted to the surface. Plot how $\mathcal{B}(z,0)$ varies with z in the range $0 \leq z \leq 5H$ for $T_s = 288\text{ K}$, $\gamma = 52\text{ K}$, $r = 1.66$, and $\tau_0 = 2.2$ (units W m^{-2} per layer). Specify what value of $\Delta z / H$ you used.
 - b) Ignoring the negligible contribution from $z > 5H$, summing over the emitted radiation that is transmitted to the surface by all layers yields the total downward flux from the atmosphere to the surface (units W m^{-2}). Calculate this value.
 - c) Imagine that the absorber mixing ratio suddenly increases and τ_0 changes from 2.2 to 2.3 but temperature remains the same. Calculate $\mathcal{B}(z,0)$ for $T_s = 288\text{ K}$, $\gamma = 52\text{ K}$, $r = 1.66$, and $\tau_0 = 2.3$. Plot how the change in flux $\mathcal{B}(z,0,\tau=2.3) - \mathcal{B}(z,0,\tau=2.2)$ varies with z in the range $0 \leq z \leq 5H$. Specify what value of $\Delta z / H$ you used.
 - d) Ignoring the negligible contribution from $z > 5H$, calculate the change in total downward flux at the surface.
 - e) Briefly explain the physical mechanism(s) responsible for the difference plot and change in total downward flux to the surface.
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10. a) Let $\mathcal{B}_0(\infty)$ be the amount of flux emitted upward by the surface that is transmitted to space. Approximating the surface as a blackbody, calculate $\mathcal{B}_0(\infty)$ if $T_s = 288\text{ K}$, $r = 1.66$, and $\tau_0 = 2.2$.
 - b) Imagine that the absorber mixing ratio suddenly increases and τ_0 changes from 2.2 to 2.3 but surface temperature remains the same. Calculate $\mathcal{B}_0(\infty)$ for $T_s = 288\text{ K}$, $r = 1.66$, and $\tau_0 = 2.3$. Calculate the change in flux $\mathcal{B}_0(\infty,\tau=2.3) - \mathcal{B}_0(\infty,\tau=2.2)$.
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11. a) Let $\mathcal{B}(z,\infty)$ be the amount of flux emitted upward by a thin atmospheric layer at height z that is transmitted to space. Plot how $\mathcal{B}(z,\infty)$ varies with z in the range $0 \leq z \leq 5H$ for $T_s = 288\text{ K}$, $\gamma = 52\text{ K}$, $r = 1.66$, and $\tau_0 = 2.2$ (units W m^{-2} per layer). Specify what value of $\Delta z / H$ you used.
 - b) Ignoring the negligible contribution from $z > 5H$, summing over the emitted radiation that is transmitted to space by all layers yields the total upward flux from the atmosphere to space (units W m^{-2}). Calculate this value.

- c) What is the total upward flux from the atmosphere and the surface to space?
- d) Imagine that the absorber mixing ratio suddenly increases and τ_0 changes from 2.2 to 2.3. Calculate $\mathcal{B}(z, \infty)$ for $T_s = 288$ K, $\gamma = 52$ K, $r = 1.66$, and $\tau_0 = 2.3$. Plot how $\mathcal{B}(z, \infty, \tau=2.3) - \mathcal{B}(z, \infty, \tau=2.2)$ varies with z in the range $0 \leq z \leq 5H$. Specify what value of $\Delta z / H$ you used.
- e) Ignoring the negligible contribution from $z > 5H$, calculate the change in total upward flux from the atmosphere to space.
- f) What is the change in the total upward flux from the atmosphere and the surface to space?
- g) Briefly explain the physical mechanism(s) responsible for the difference plot and change in total upward flux from the atmosphere and surface to space.

Note that the level of maximum flux transmitted to space in 11.a) occurs at a lower height than would be calculated from the answer to 5.e). This is because temperature decreases with height in 11.a), thus giving more weight to radiation emitted lower in the atmosphere. If you do not understand how this is so, come talk to me.